

Модуль 4. Лабораторная 5

1) Для заданной на схеме schema-lab5 сети, состоящей из управляемых коммутаторов, маршрутизаторов и персональных компьютеров выполнить планирование и документирование адресного пространства и назначить статические адреса всем устройствам. nb! Каждое соединение маршрутизатора с маршрутизатором — это отдельная сеть.

Поднимаем интерфейс на R1:

```
configure terminal

interface FastEthernet0/0
no shutdown
exit

interface FastEthernet1/0
no shutdown
exit

end
```

Поднимаем интерфейс на R6:

```
configure terminal

interface FastEthernet0/0
no shutdown
exit

interface FastEthernet1/0
no shutdown
exit

interface FastEthernet2/0
no shutdown
exit

end
```

Настраиваем R1:

```
configure terminal

interface FastEthernet0/0
ip address 192.168.51.1 255.255.255.0
exit

interface FastEthernet1/0
ip address 192.168.123.1 255.255.255.0
exit

end
```

Настраиваем R6:

```
configure terminal

interface FastEthernet0/0
```

```

ip address 192.168.26.6 255.255.255.0
exit

interface FastEthernet1/0
ip address 192.168.3.1 255.255.255.0
exit

interface FastEthernet2/0
ip address 192.168.67.6 255.255.255.0
exit

end

```

Настраиваем PC4:

```

ip 192.168.4.2/24 192.168.4.1
save

```

Настраиваем коммутатор:

```

configure terminal

interface Vlan1
ip address 192.168.123.10 255.255.255.0
no shutdown
exit

ip default-gateway 192.168.123.1

end

```

Устройство	Интерфейс	IP-адрес	Маска	Назначение / сосед
PC1	Ethernet	192.168.1.2	/24	сеть с R4
R4	Fa0/0	192.168.1.1	/24	PC1
R4	Fa1/0	192.168.45.4	/24	R5
R5	Fa1/0	192.168.45.5	/24	R4
PC2	Ethernet	192.168.2.2	/24	сеть с R5
R5	Fa0/0	192.168.2.1	/24	PC2
R5	Fa2/0	192.168.51.5	/24	R1
R1	Fa0/0	192.168.51.1	/24	R5
R1	Fa1/0	192.168.123.1	/24	Коммутатор
R2	Fa0/0	192.168.123.2	/24	Коммутатор
R3	Fa0/0	192.168.123.3	/24	Коммутатор
Layer2Switch-1	Vlan1	192.168.123.10	/24	управляющий IP
R2	Fa1/0	192.168.26.2	/24	R6
R6	Fa0/0	192.168.26.6	/24	R2
PC3	Ethernet	192.168.3.2	/24	сеть с R6
R6	Fa1/0	192.168.3.1	/24	PC3
R6	Fa2/0	192.168.67.6	/24	R7
R7	Fa2/0	192.168.67.7	/24	R6
R3	Fa1/0	192.168.37.3	/24	R7
R7	Fa0/0	192.168.37.7	/24	R3
PC4	Ethernet	192.168.4.2	/24	сеть с R7
R7	Fa1/0	192.168.4.1	/24	PC4
R3	Fa2/0	192.168.38.3	/24	R8
R8	Fa0/0	192.168.38.8	/24	R3
PC5	Ethernet	192.168.5.2	/24	сеть с R8
R8	Fa1/0	192.168.5.1	/24	PC5

Шлюзы по умолчанию для ПК:

PC1 - 192.168.1.1

PC2 - 192.168.2.1

PC3 - 192.168.3.1

PC4 - 192.168.4.1

PC5 - 192.168.5.1

Список сетей:

192.168.1.0/24 PC1 — R4

192.168.2.0/24 PC2 — R5

192.168.3.0/24 PC3 — R6

192.168.4.0/24 PC4 — R7

192.168.5.0/24 PC5 — R8

192.168.45.0/24 R4 — R5

192.168.51.0/24 R5 — R1

192.168.123.0/24 R1 — R2 — R3 через Switch

192.168.26.0/24 R2 — R6

192.168.37.0/24 R3 — R7

192.168.38.0/24 R3 — R8

2) Настроить протокол динамической маршрутизации RIP v2 для области, указанной на схеме schema-lab5.

Настраиваем RIP v2 на R4:

```
configure terminal

router rip
version 2
no auto-summary
network 192.168.1.0
network 192.168.45.0
passive-interface FastEthernet0/0

end
```

Настраиваем RIP v2 на R5:

```
configure terminal

router rip
version 2
no auto-summary
network 192.168.2.0
```

```
network 192.168.45.0
network 192.168.51.0
passive-interface FastEthernet0/0

end
```

Настраиваем RIP v2 на R1:

```
configure terminal

router rip
version 2
no auto-summary
network 192.168.51.0

end
```

```
R1#show ip route rip
R   192.168.45.0/24 [120/1] via 192.168.51.5, 00:00:06, FastEthernet0/0
R   192.168.1.0/24 [120/2] via 192.168.51.5, 00:00:06, FastEthernet0/0
R   192.168.2.0/24 [120/1] via 192.168.51.5, 00:00:06, FastEthernet0/0
R1#
```

```
R4#show ip route rip
R   192.168.51.0/24 [120/1] via 192.168.45.5, 00:00:08, FastEthernet1/0
R   192.168.2.0/24 [120/1] via 192.168.45.5, 00:00:08, FastEthernet1/0
R4#
```

```
R5#show ip route rip
R   192.168.1.0/24 [120/1] via 192.168.45.4, 00:00:17, FastEthernet1/0
R5#
```

- 3) Настроить протокол динамической маршрутизации OSPF для зон 0, 1,
2. Зону 1 настроить как полностью (nb!) тупиковую.

Настраиваем OSPF на R1:

```
configure terminal

router ospf 1
network 192.168.123.0 0.0.0.255 area 0

end
```

Настраиваем OSPF на R2:

```
configure terminal

router ospf 1
network 192.168.123.0 0.0.0.255 area 0
network 192.168.26.0 0.0.0.255 area 2

end
```

Настраиваем OSPF на R3:

```
configure terminal

router ospf 1
network 192.168.123.0 0.0.0.255 area 0
```

```
network 192.168.37.0 0.0.0.255 area 2
network 192.168.38.0 0.0.0.255 area 1
```

```
end
```

Настраиваем OSPF на R8:

```
configure terminal

router ospf 1
network 192.168.38.0 0.0.0.255 area 1
network 192.168.5.0 0.0.0.255 area 1
passive-interface FastEthernet1/0

end
```

Начинаем превращать Area 1 в полностью тупиковую:

Настраиваем R8:

```
configure terminal

router ospf 1
area 1 stub

end
```

R3:

```
configure terminal

router ospf 1
area 1 stub no-summary

end
```

Настраиваем Area 2 на R6:

```
configure terminal

router ospf 1
network 192.168.26.0 0.0.0.255 area 2
network 192.168.67.0 0.0.0.255 area 2
network 192.168.3.0 0.0.0.255 area 2
passive-interface FastEthernet1/0

end
```

Настраиваем OSPF на R7:

```
configure terminal

router ospf 1
network 192.168.37.0 0.0.0.255 area 2
network 192.168.67.0 0.0.0.255 area 2
network 192.168.4.0 0.0.0.255 area 2
passive-interface FastEthernet1/0

end
```

4) Настроить редистрибуцию маршрутов между протоколами RIP v2 и OSPF.

R1:

```
configure terminal

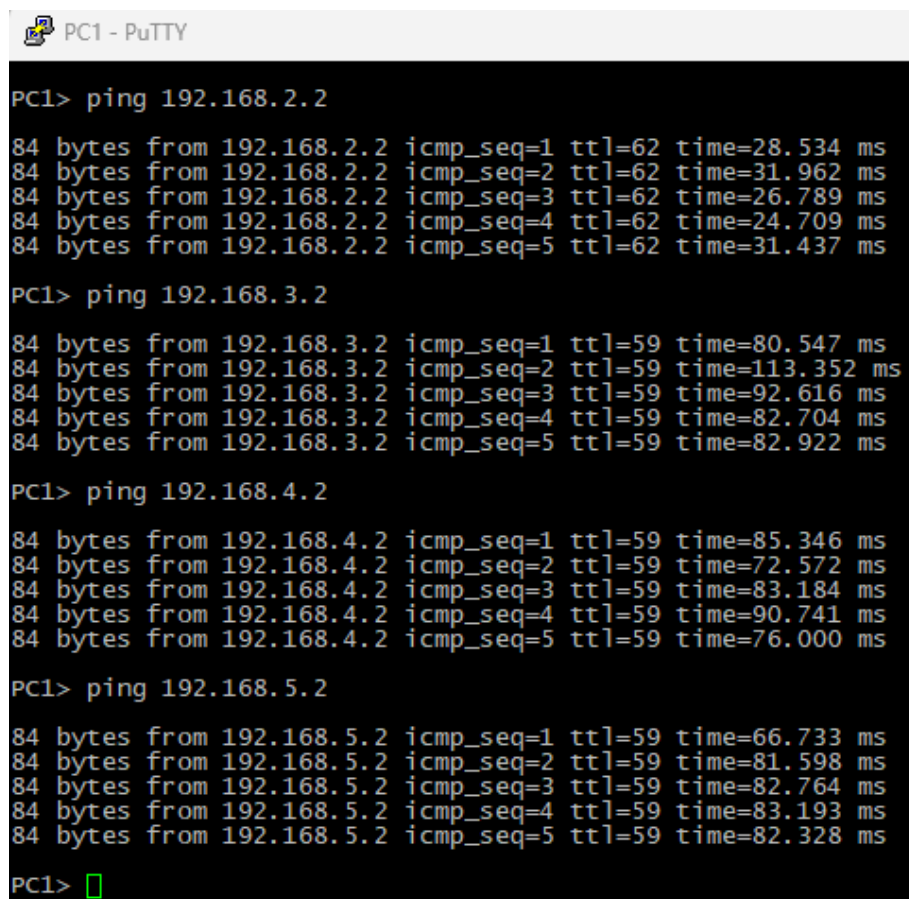
router ospf 1
 redistribute rip subnets

router rip
 redistribute ospf 1 metric 2

end
```

5) Проверить работоспособность маршрутизации, выполнив ping VPC "все между всеми" (nb!: в обе стороны).

PC1:



PC1 - PuTTY

```
PC1> ping 192.168.2.2

84 bytes from 192.168.2.2 icmp_seq=1 ttl=62 time=28.534 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=62 time=31.962 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=62 time=26.789 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=62 time=24.709 ms
84 bytes from 192.168.2.2 icmp_seq=5 ttl=62 time=31.437 ms

PC1> ping 192.168.3.2

84 bytes from 192.168.3.2 icmp_seq=1 ttl=59 time=80.547 ms
84 bytes from 192.168.3.2 icmp_seq=2 ttl=59 time=113.352 ms
84 bytes from 192.168.3.2 icmp_seq=3 ttl=59 time=92.616 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=59 time=82.704 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=59 time=82.922 ms

PC1> ping 192.168.4.2

84 bytes from 192.168.4.2 icmp_seq=1 ttl=59 time=85.346 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=59 time=72.572 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=59 time=83.184 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=59 time=90.741 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=59 time=76.000 ms

PC1> ping 192.168.5.2

84 bytes from 192.168.5.2 icmp_seq=1 ttl=59 time=66.733 ms
84 bytes from 192.168.5.2 icmp_seq=2 ttl=59 time=81.598 ms
84 bytes from 192.168.5.2 icmp_seq=3 ttl=59 time=82.764 ms
84 bytes from 192.168.5.2 icmp_seq=4 ttl=59 time=83.193 ms
84 bytes from 192.168.5.2 icmp_seq=5 ttl=59 time=82.328 ms

PC1> █
```

PC2:

```
PC2 - PuTTY

PC2> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=62 time=28.967 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=62 time=32.418 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=62 time=25.252 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=62 time=22.277 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=62 time=41.641 ms

PC2> ping 192.168.3.2

84 bytes from 192.168.3.2 icmp_seq=1 ttl=60 time=71.030 ms
84 bytes from 192.168.3.2 icmp_seq=2 ttl=60 time=62.806 ms
84 bytes from 192.168.3.2 icmp_seq=3 ttl=60 time=63.815 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=60 time=83.677 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=60 time=74.196 ms

PC2> ping 192.168.4.2

84 bytes from 192.168.4.2 icmp_seq=1 ttl=60 time=64.370 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=60 time=88.943 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=60 time=61.621 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=60 time=96.931 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=60 time=77.242 ms

PC2> ping 192.168.5.2

84 bytes from 192.168.5.2 icmp_seq=1 ttl=60 time=62.205 ms
84 bytes from 192.168.5.2 icmp_seq=2 ttl=60 time=65.742 ms
84 bytes from 192.168.5.2 icmp_seq=3 ttl=60 time=72.436 ms
84 bytes from 192.168.5.2 icmp_seq=4 ttl=60 time=75.877 ms
84 bytes from 192.168.5.2 icmp_seq=5 ttl=60 time=76.858 ms

PC2> █
```

PC3:

```
PC3 - PuTTY

PC3> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=59 time=100.688 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=59 time=63.788 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=59 time=83.911 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=59 time=78.383 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=59 time=87.757 ms

PC3> ping 192.168.2.2

84 bytes from 192.168.2.2 icmp_seq=1 ttl=60 time=77.488 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=60 time=79.923 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=60 time=55.505 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=60 time=66.684 ms
84 bytes from 192.168.2.2 icmp_seq=5 ttl=60 time=65.828 ms

PC3> ping 192.168.4.2

84 bytes from 192.168.4.2 icmp_seq=1 ttl=62 time=44.719 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=62 time=22.588 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=62 time=43.318 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=62 time=32.608 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=62 time=30.217 ms

PC3> ping 192.168.5.2

84 bytes from 192.168.5.2 icmp_seq=1 ttl=60 time=72.203 ms
84 bytes from 192.168.5.2 icmp_seq=2 ttl=60 time=61.251 ms
84 bytes from 192.168.5.2 icmp_seq=3 ttl=60 time=44.557 ms
84 bytes from 192.168.5.2 icmp_seq=4 ttl=60 time=65.910 ms
84 bytes from 192.168.5.2 icmp_seq=5 ttl=60 time=63.352 ms

PC3>
PC3> █
```

PC4:

```
PC4 - PuTTY

PC4> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=59 time=71.465 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=59 time=84.771 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=59 time=76.302 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=59 time=85.636 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=59 time=95.380 ms

PC4> ping 192.168.2.2

84 bytes from 192.168.2.2 icmp_seq=1 ttl=60 time=62.026 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=60 time=75.245 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=60 time=74.155 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=60 time=64.296 ms
84 bytes from 192.168.2.2 icmp_seq=5 ttl=60 time=82.581 ms

PC4> ping 192.168.3.2

84 bytes from 192.168.3.2 icmp_seq=1 ttl=62 time=39.613 ms
84 bytes from 192.168.3.2 icmp_seq=2 ttl=62 time=31.821 ms
84 bytes from 192.168.3.2 icmp_seq=3 ttl=62 time=23.258 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=62 time=23.575 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=62 time=32.634 ms

PC4> ping 192.168.5.2

84 bytes from 192.168.5.2 icmp_seq=1 ttl=61 time=44.003 ms
84 bytes from 192.168.5.2 icmp_seq=2 ttl=61 time=51.113 ms
84 bytes from 192.168.5.2 icmp_seq=3 ttl=61 time=55.233 ms
84 bytes from 192.168.5.2 icmp_seq=4 ttl=61 time=51.896 ms
84 bytes from 192.168.5.2 icmp_seq=5 ttl=61 time=32.631 ms

PC4>
PC4> █
```

PC5:

```
PC5 - PuTTY

PC5> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=59 time=103.919 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=59 time=66.881 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=59 time=76.264 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=59 time=90.886 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=59 time=73.858 ms

PC5> ping 192.168.2.2

84 bytes from 192.168.2.2 icmp_seq=1 ttl=60 time=80.325 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=60 time=70.444 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=60 time=62.517 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=60 time=67.817 ms
84 bytes from 192.168.2.2 icmp_seq=5 ttl=60 time=79.347 ms

PC5> ping 192.168.3.2

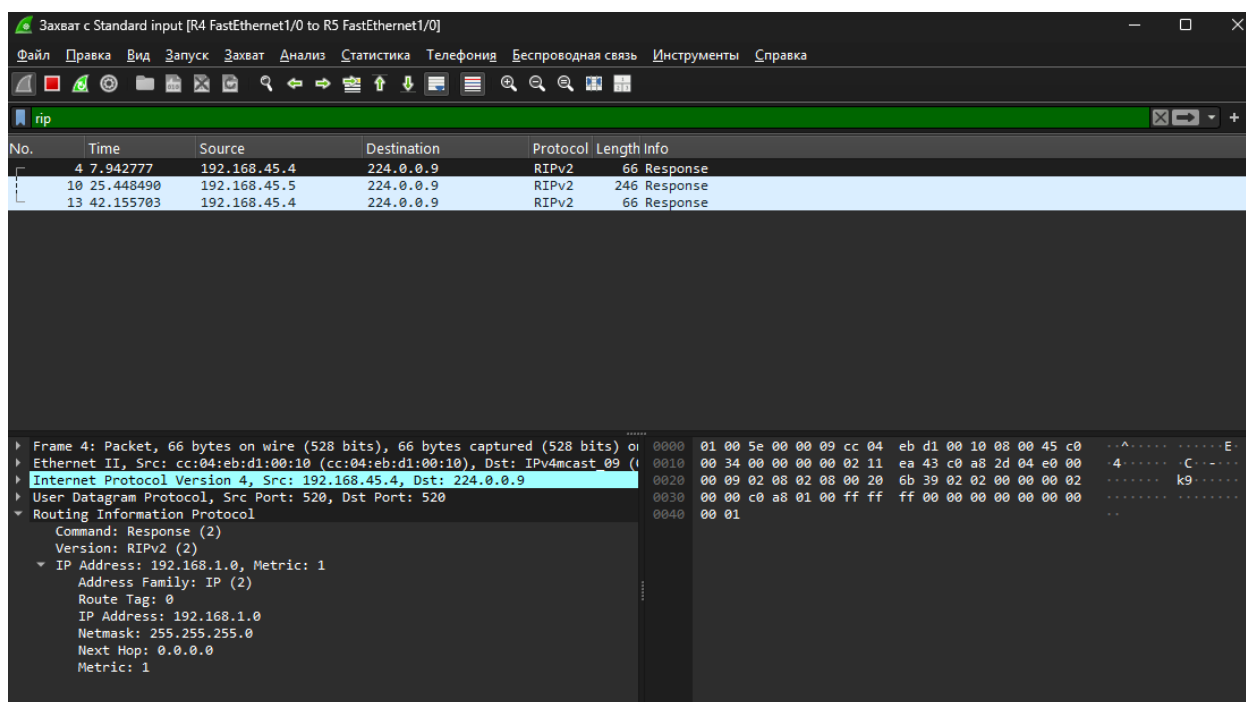
84 bytes from 192.168.3.2 icmp_seq=1 ttl=60 time=73.405 ms
84 bytes from 192.168.3.2 icmp_seq=2 ttl=60 time=52.881 ms
84 bytes from 192.168.3.2 icmp_seq=3 ttl=60 time=51.391 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=60 time=45.842 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=60 time=53.905 ms

PC5> ping 192.168.4.2

84 bytes from 192.168.4.2 icmp_seq=1 ttl=61 time=61.277 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=61 time=50.916 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=61 time=45.446 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=61 time=35.428 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=61 time=47.332 ms

PC5>
PC5> █
```


6) Перехватить в Wireshark сообщения протоколов RIP v2 и OSPF, идентифицировать их тип и содержание.



Пакет протокола RIP v2 на соединении R4–R5. Пакет отправлен маршрутизатором R4 с адреса 192.168.45.4 на multicast-адрес 224.0.0.9. Тип сообщения — Response, версия протокола — RIPv2. В сообщении передаётся маршрут к сети 192.168.1.0/24 с метрикой 1. В качестве маски сети указана 255.255.255.0, поле Next Hop имеет значение 0.0.0.0. Для передачи RIP используется UDP-порт 520.

*Standard input [R1 FastEthernet1/0 to Layer2Switch-1 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

ospf

No.	Time	Source	Destination	Protocol	Length	Info
3	1.364830	192.168.123.1	224.0.0.5	OSPF	98	Hello Packet
4	2.659645	192.168.123.2	224.0.0.5	OSPF	98	Hello Packet
8	6.970175	192.168.123.3	224.0.0.5	OSPF	98	Hello Packet
11	11.373420	192.168.123.1	224.0.0.5	OSPF	98	Hello Packet
12	12.636399	192.168.123.2	224.0.0.5	OSPF	98	Hello Packet
13	12.758440	192.168.123.3	224.0.0.6	OSPF	90	LS Update
14	12.765108	192.168.123.1	224.0.0.5	OSPF	90	LS Update
15	12.789785	192.168.123.3	224.0.0.6	OSPF	90	LS Update
16	12.806153	192.168.123.1	224.0.0.5	OSPF	90	LS Update
19	15.301576	192.168.123.2	224.0.0.5	OSPF	98	LS Acknowledge
21	16.956281	192.168.123.3	224.0.0.5	OSPF	98	Hello Packet
24	21.359655	192.168.123.1	224.0.0.5	OSPF	98	Hello Packet
25	22.632986	192.168.123.2	224.0.0.5	OSPF	98	Hello Packet

Frame 3: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on Ethernet II, Src: cc:01:eb:7d:00:10 (cc:01:eb:7d:00:10), Dst: IPv4mcast_05 (01:00:5e:00:00:05)

Internet Protocol Version 4, Src: 192.168.123.1, Dst: 224.0.0.5

Open Shortest Path First

- OSPF Header
 - Version: 2
 - Message Type: Hello Packet (1)
 - Packet Length: 52
 - Source OSPF Router: 192.168.123.1
 - Area ID: 0.0.0.0 (Backbone)
 - Checksum: 0xc23f [correct]
 - Instance ID: Base IPv4 Unicast Instance (0)
 - Auth Type: Null (0)
 - Auth Data (none): 0000000000000000
- OSPF Hello Packet
 - Network Mask: 255.255.255.0
 - Hello Interval [sec]: 10
 - Options: 0x12, (L) LLS Data block, (E) External Routing
 - Router Priority: 1
 - Router Dead Interval [sec]: 40
 - Designated Router: 192.168.123.1
 - Backup Designated Router: 192.168.123.2
 - Active Neighbor: 192.168.123.2
 - Active Neighbor: 192.168.123.3
- OSPF LLS Data Block

Пакет OSPFv2 типа Hello. Пакет отправлен маршрутизатором R1 с адреса 192.168.123.1 на multicast-адрес 224.0.0.5. Пакет относится к Backbone Area 0. В Hello-сообщении передаются параметры соседства OSPF: маска сети 255.255.255.0, Hello Interval 10 с, Dead Interval 40 с, приоритет маршрутизатора, адреса DR и BDR, а также список активных соседей. В данном сегменте R1 является DR, R2 — BDR, а R2 и R3 указаны как активные соседи.

Standard input [R1 FastEthernet1/0 to Layer2Switch-1 Ethernet0]

Файл Плавка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

ospf

No.	Time	Source	Destination	Protocol	Length	Info
3	1.364830	192.168.123.1	224.0.0.5	OSPF	98	Hello Packet
4	2.659645	192.168.123.2	224.0.0.5	OSPF	98	Hello Packet
8	6.970175	192.168.123.3	224.0.0.5	OSPF	98	Hello Packet
11	11.373420	192.168.123.1	224.0.0.5	OSPF	98	Hello Packet
12	12.636399	192.168.123.2	224.0.0.5	OSPF	98	Hello Packet
13	12.758440	192.168.123.3	224.0.0.6	OSPF	90	LS Update
14	12.765108	192.168.123.1	224.0.0.5	OSPF	90	LS Update
15	12.789785	192.168.123.3	224.0.0.6	OSPF	90	LS Update
16	12.806153	192.168.123.1	224.0.0.5	OSPF	90	LS Update
19	15.301576	192.168.123.2	224.0.0.5	OSPF	98	LS Acknowledge
21	16.956281	192.168.123.3	224.0.0.5	OSPF	98	Hello Packet
24	21.359655	192.168.123.1	224.0.0.5	OSPF	98	Hello Packet
25	22.632986	192.168.123.2	224.0.0.5	OSPF	98	Hello Packet

Frame 13: Packet, 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on 0

Ethernet II, Src: cc:03:eb:b5:00:00 (cc:03:eb:b5:00:00), Dst: IPv4mcast_06 (01:00:00:00:00:06)

Internet Protocol Version 4, Src: 192.168.123.3, Dst: 224.0.0.6

Open Shortest Path First

- OSPF Header
 - Version: 2
 - Message Type: LS Update (4)
 - Packet Length: 56
 - Source OSPF Router: 192.168.123.3
 - Area ID: 0.0.0.0 (Backbone)
 - Checksum: 0xecea [correct]
 - Instance ID: Base IPv4 Unicast Instance (0)
 - Auth Type: Null (0)
 - Auth Data (none): 0000000000000000
- LS Update Packet
 - Number of LSAs: 1
 - LSA-type 3 (Summary-LSA (IP network)), len 28
 - .000 0000 0000 0001 = LS Age (seconds): 1
 - 0... .. = Do Not Age Flag: 0
 - Options: 0x22, (DC) Demand Circuits, (E) External Routing
 - LS Type: Summary-LSA (IP network) (3)
 - Link State ID: 192.168.5.0
 - Advertising Router: 192.168.123.3
 - Sequence Number: 0x80000003
 - Checksum: 0x32b0
 - Length: 28
 - Netmask: 255.255.255.0
 - TOS: 0
 - Metric: 2

Пакет OSPFv2 типа LS Update. Пакет отправлен маршрутизатором R3 с адреса 192.168.123.3 на multicast-адрес 224.0.0.6. Пакет относится к Backbone Area 0. Внутри содержится одна LSA типа Summary-LSA (Type 3), объявляющая сеть 192.168.5.0/24. В качестве Advertising Router указан R3 (192.168.123.3), а метрика маршрута равна 2. Данный тип LSA используется ABR для распространения информации о сетях из одной OSPF-зоны в другую.